



OPEN Melt-driven freshwater inputs as potential drivers of harmful algal blooms: evidence from a diatom bloom in Northwestern Patagonia

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Harmful algal blooms (HABs) in coastal waters are an increasing ecological and socio-economic concern in Northwestern Patagonia (NWP). One widely accepted mechanism proposes that reduced freshwater input weakens water-column stratification and increases residence time, allowing nutrient-rich oceanic waters to intrude and favor HAB development. Here, we demonstrate that riverine controls exerted on HABs extend beyond low-flow conditions to include shifts in the timing and magnitude of high-flow pulses. Using long-term hydro-meteorological records, satellite-based data, and a citizen science water quality time series from the Puelo River, we assess how peak freshwater inputs modulated the conditions leading up to a diatom HAB (*Thalassiosira cf. pseudonana*) in the Reloncaví Fjord (41.5°S) in November 2023. Based on inference from spring 2022 and 2024 monitoring, such discharge peaks provide short-lived windows of enhanced nutrient availability, characterized by elevated reactive silica ($> 100 \mu\text{mol L}^{-1}$), dissolved iron ($> 0.3 \mu\text{mol L}^{-1}$) and nitrogen-rich nutrient ratios ($\text{N:P} > 16$). An anomalous high-turbidity pulse, usually dampened by downstream lake regulation, suggests threshold effect of meltwater inputs, which likely gave this taxon a competitive advantage. Antecedent conditions also include a sustained winter runoff effect on stratification and exchange flow, followed by an unusually dry early spring, weakening stratification during the period of increased solar radiation, both preconditioning the system for the HAB.

Keywords Freshwater input, Harmful algal blooms, Diatoms, Aquaculture, Climate change, Fjords, Northwestern patagonia

River regimes regulate the retention and transport/export of materials, nutrients, and other resources downstream, hence these processes are defined by a wide array of natural fluvial processes such as floods, droughts, and erosion. These regimes are further modified by anthropogenic interventions such as dams, land use change, and climate change^{1–3}. Consequently, the intensification and spatial expansion of anthropogenic pressures have drastically reduced opportunities to understand river systems under reference conditions⁴.

Northwestern Patagonia (NWP, 41–46.5° S; Fig. 1A) is one of the few regions worldwide where rivers with minimal human interference^{5,6} can be studied. Its scientific relevance is amplified by the fact that during the past 7 decades, NWP has experienced a sustained decline in precipitation during autumn ($-8 \pm 8\%$ per decade;

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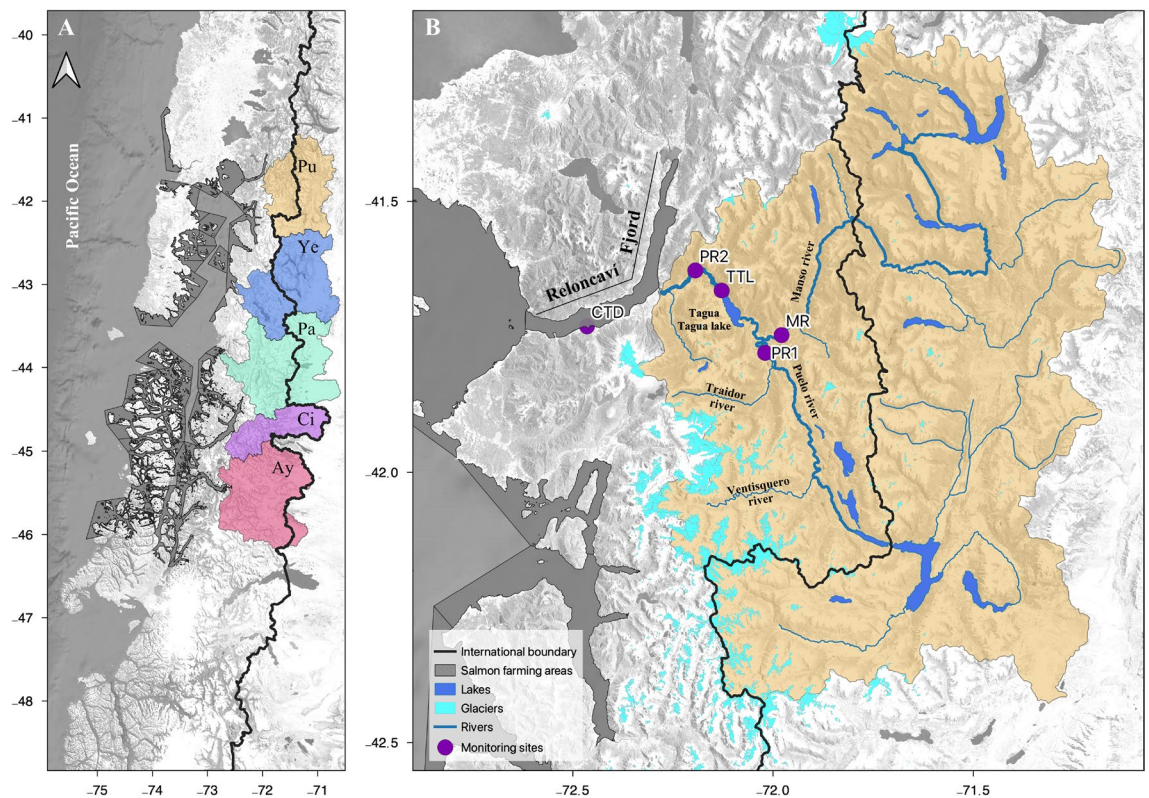


Fig. 1. (A) Puelo River (Pu) basin in the context of the major basins of Northwestern Patagonia (NWP): Yelcho (Ye), Palena (Pa), Cisnes (Ci), and Aysén (Ay) Rivers. The black line delineates the Chile–Argentina international boundary. Shaded marine zones indicate areas authorized for salmon farming. (B) Detailed view of the Puelo River basin and the Reloncaví Fjord. The purple circles indicate the monitoring stations: Manso River (MR); Puelo River, upstream of MR (PR1); Tagua Tagua Lake (TTL); Puelo River, in the final section (PR2, Carrera Basilio gauging station); and inside the Reloncaví Fjord (CTD). The figure was produced using QGIS v3.28.0-Firenze, with Bing Maps aerial imagery (<https://www.bing.com/maps>) used as the background basemap.

$p < 0.01$) and an increasing frequency of droughts^{7–10}. These climatic trends have been partly attributed to a positive trend in the Southern Annular Mode (SAM), linked to stratospheric ozone depletion and increased greenhouse gas concentrations¹¹. In addition, the combined influence of a positive SAM phase and the warm phases of El Niño–Southern Oscillation (ENSO) has been particularly disruptive in NWP, driving anomalously dry periods across the region¹². As a result, the decline in precipitation has altered the hydrological signature of freshwater inputs to one of South America's largest fjords and inner sea systems¹³ (Fig. 1). Climate projections suggest that patterns may become even more pronounced towards the end of the century¹⁴. A key sentinel of these changes is the Puelo River, one of the largest rivers in NWP and the main freshwater input to the Reloncaví fjord^{14,15} (Fig. 1B). Analyses of its long-term discharge records have revealed alterations in the timing, duration, and frequency of minimum and maximum hydrological events in the Puelo River, including a clear declining trend in streamflow, particularly during summer and autumn^{14–16} (e.g., May: -9% per decade; $p < 0.05$).

River discharge is the main source of density variation in the inner sea and plays a key role in shaping water-mass structure in NWP fjords. In these systems, the interaction between freshwater inputs and oceanic water masses, primarily the high-salinity Subantarctic Water (SAAW), generates strong salinity stratification, which, together with other forcing mechanisms (e.g., wind and tides), regulates circulation patterns, water renewal and ventilation^{17,18}. Recent hydroclimatic changes are already altering these processes. Reductions and shifts in freshwater input across the NWP over recent decades have led to a shallower halocline and enhanced surface warming trends in key coastal systems, such as the Reloncaví Fjord and Sound¹⁹ (Fig. 1).

Beyond their physical influence on stratification and circulation, riverine inputs also exert strong control over the biogeochemical functioning of fjords and inner sea systems in NWP through multiple pathways: (i) suspended sediment inputs from glacial meltwater or soil erosion, may influence marine productivity by regulating the light regime²⁰ or supplying substrate²¹; (ii) inorganic macronutrients (NO_3^- , NH_4^+ , PO_4^{3-}) play a crucial role in limiting primary productivity within ecosystems, while excessive loading may drive tipping points in marine ecosystems at large scales^{22,23}; (iii) diatoms require silicic acid (dissolved silica - dSi), while dissolved iron (dFe) is a key micronutrient regulating phytoplankton growth more broadly; both may be provided principally by riverine inputs, often at concentrations one to two orders of magnitude higher than those in the marine environment^{24,25}; and (iv) organic carbon exported from rivers provides energy, through

Particulate Organic Matter (POM) inputs to benthic communities²¹, and Dissolved Organic Carbon (DOC) contribution to microbial production.

The hydrological and biogeochemical signatures of rivers also have far-reaching implications for economic activities that depend on the land-sea interface²⁶. NWP exemplifies this dynamic, as its macro-estuarine system supports a range of socio-economic activities, with salmon farming being one of the most significant²⁷. Chile is the world's second-largest exporter of salmonids, with an annual harvest of 1.09 million tons in 2023 and export revenues totaling US\$6.46 billion²⁸. Each year, Chilean salmon farming takes place in floating cages located in about 400 marine concession sites distributed throughout the complex coastal region of Western Patagonia²⁹. In 2024, 88% of Chile's total salmon and trout harvest originated from the NWP, with nearly half of that total coming from fjords and connected areas with medium to high freshwater influence^{19,30}. As salmon farming is highly dependent on freshwater influence, it is also highly exposed (biomass loss) to extreme hydroclimatic events^{27,30}. According to Chile's National Fisheries and Aquaculture Service (SERNAPESCA), environmental factors represent the second most prevalent cause of mortality in salmon and trout, with harmful algal blooms (HABs) constituting the primary extreme event triggering these losses³⁰. This vulnerability underscores the potential relevance of the tight coupling between hydrological variability and the biological state of the fjord ecosystems.

Emerging evidence indicates that climate change is already reshaping the composition, frequency, and severity of HABs in marine and brackish systems^{31,32}. However, in NWP, understanding the mechanisms that trigger HABs remains a significant scientific challenge due to the strong coupling between physical, chemical, and ecological processes³³, as well as the contrasting responses of different functional groups to hydrological and oceanographic conditions³⁴. A striking example of this occurred during the 2016 austral summer–autumn, when a positive SAM phase, combined with El Niño conditions, led to one of the most severe droughts on record in the NWP. During this event, exceeded 50% between 40° and 47° S, while streamflow anomalies ranged from −30% to −60% relative to long-term means^{12,35}. Under these conditions, a massive HAB of the ichthyotoxic dictyochophyte *Pseudochattonella verruculosa* resulted in the loss of over 40,000 tonnes of salmon (~US\$800 million), supported by its capacity to grow at intermediate salinities and accumulate maximum biomass at moderately high salinity (~30 psu)³⁶. Within just few days, this HAB event impacted aquaculture and fisheries across multiple coastal systems, including the central–northern inland sea of Chiloé as well as the Reloncaví Sound and Fjord (Fig. 1)^{35,37}. In major HAB events such as the 2016 outbreak, hydrological shifts and their influence on estuarine stratification have been identified as key drivers due to reductions in freshwater input, as a primary physical control on their occurrence and persistence^{30,33,38}. The most widely accepted mechanism proposes that reduced freshwater input weakens water-column stratification and increases residence time, thereby allowing the intrusion of nutrient-rich oceanic waters that promote bloom development^{15,35,37}. Such conditions are typically associated with flagellate blooms emerging during prolonged stability and low-flow anomalies. However, other HAB-forming taxa (e.g., *Pseudo-nitzschia*, *Dinophysis* and *Karenia*) may respond to very different hydrological cues. Moreover, low-flow conditions often coincide with dry periods characterized by enhanced solar radiation and elevated surface water temperatures³⁹.

Despite this growing body of evidence, the role of hydrological forcing in shaping HAB dynamics in Chile remains incompletely resolved, particularly with respect to freshwater variability and high-flow pulses. Increasing evidence further suggests that hydrological influence on HABs is not unidirectional, thus rapid freshwater pulses can also restructure the surface layer in ways that favor fast-growing diatoms, as observed during the 2014–2015 North Pacific marine heatwave, when elevated rainfall in inner fjords enhanced diatom growth in the upper layer⁴⁰. Diatom blooms in Chilean fjords exemplify this gap. *Thalassiosira cf. pseudonana*, a small centric taxon responsible for salmon mortality events in the region since 2014⁴¹, was not formally identified in Chilean waters until strains isolated from recent bloom events were characterized genetically and morphologically⁴².

Given the highly dynamic nature of river forcing and its sensitivity to climate-driven changes in freshwater inputs, this study explores underexamined mechanisms linking river and fjord ecosystems and influencing HAB occurrence in NWP. Specifically, we evaluate the link between the timing and magnitude of meltwater-driven inputs from Puelo River and the development of HABs in the Reloncaví Fjord (41.5°S). Accordingly, we hypothesize that an otherwise normal spring melt-driven freshwater input, preceded by intra- and/or interannual dry periods, generate physical and biogeochemical shifts in the coastal water column, enhancing short-term growth conditions (nutrients, light, salinity) that facilitate diatom HABs in strongly stratified fjords.

Methods

Study area

The Reloncaví Fjord (Fig. 1B) is a 55 km long J-shaped estuarine system which narrows to less than 3 km at its head, reaching depths of over 400 m near its mouth. This fjord is strongly influenced by substantial freshwater inputs, which generate a three-layer circulation system⁴³. The low-salinity surface layer flows seaward from head to mouth, supporting key economic activities such as salmon farming and wild mussel seed collection for mussel farming^{30,44}. Between 2016 and 2022, an average of approximately $30,000 \pm 14,300$ tons of salmonid, primarily rainbow trout (*Oncorhynchus mykiss*) and Pacific salmon (*Oncorhynchus kisutch*), were harvested annually by salmon farmers, across an average of 15 ± 4 aquaculture concessions located on both sides of the fjord. The main tributary of the Reloncaví fjord is the Puelo River (~9,000 km², Fig. 1), which discharges into the middle section of the fjord with a mean flow of 630 m³ s^{−1}. The Puelo River basin landscape is dominated by native forests with minimal human intervention (24% of the basin is in protected areas). Precipitation occurs year-round, with higher frequency during the austral winter, and exhibits a pronounced west–east gradient, ranging from ~4,700 mm on the western side to about ~500 mm in the eastern (Argentine) sector⁴⁵. This gradient results from orographic precipitation enhancement on the western flank of the Austral Andes and a rain-shadow effect toward the east. On average, the Puelo River exhibits a pluvial–nival hydrological regime, characterized

by a bimodal seasonal discharge pattern with a primary peak during the austral winter driven by rainfall and a secondary peak in spring associated with snowmelt¹⁶.

Data

To analyze the link between melt-driven freshwater inputs and HAB, we integrated datasets spanning multiple temporal scales (Table 1). Long-term hydroclimatic records were used to establish baseline conditions and characterize variability, whereas high-frequency and event-based datasets were used to resolve conditions before, during and after HABs.

Harmful algal blooms dynamics

In this study, HAB events were identified using three criteria:

- (i) The presence of potentially harmful taxa, assessed using data from the Phytoplankton Monitoring Program (PROMOFI), a long-term initiative established in 1988 to monitor and mitigate HAB impacts on Chilean salmon farming. PROMOFI provides information on sampling date, site, depth, species composition, and cell concentrations (cells mL⁻¹), with increased sampling frequency during confirmed events.
- (ii) The exceedance of species-specific risk thresholds defined by SERNAPESCA (Resolution DN-N°00371/2024). A HAB Index was calculated as the ratio between observed cell concentrations and these thresholds, where values > 0 indicate exceedance of critical levels.
- (iii) The temporal correspondence between phytoplankton observations and salmon mortality records (2012–2024), reported to SERNAPESCA by aquaculture companies using concession-level identifiers, including information on species, number of affected fish, average weight, and attributed causes of death (e.g., HAB, hypoxia, disease, or management-related discard).

Meteorological conditions

Daily precipitation and air temperature were obtained from the ERA5 reanalysis⁴⁶(1960–2024; 0.25°). ERA5 was selected because it is widely used in the study area⁴⁵ and has demonstrated robust performance in capturing both the magnitude and temporal variability of key meteorological drivers, while also offering near-real-time availability that supports timely assessment of hydroclimatic conditions. The gridded values were aggregated at the catchment scale for the Puelo River Basin.

Freshwater input

To characterize how variability in freshwater inputs propagates into physicochemical conditions relevant for fjord ecosystems, we combined (i) hydrological records with (ii) in situ monitoring, (iii) targeted water-sampling campaigns, and (iv) satellite-derived data:

- i) Hydrological records: River flow variability was quantified using daily discharge records from the Puelo River (1950–2024, Carrera Basilio gauging station), located 12 km upstream of the Reloncaví Fjord and maintained by the Chilean General Water Directorate (DGA) (Fig. 1, PR2).
- ii) In situ monitoring: Freshwater-driven physicochemical signals were then tracked along the river–fjord continuum using continuous sensor measurements. Between December 2022 and January 2025, we recorded 10-minute time series of turbidity, and temperature at a depth of 2 m, adjacent to the Carrera Basilio gauging station (Fig. 1). Turbidity was measured using a Turner Designs Cyclops-7 Logger sensor, while dissolved oxygen and temperature were recorded using a PME MiniDOT. Both instruments were equipped with wiper systems and programmed to record at 10-minute intervals. The water temperature dataset was complemented with the long-term hourly temperature recorded by the DGA gauging stations at Carrera Basilio, Tagua Tagua Lake, the Manso River (MR), and the Puelo River upstream of the MR (Fig. 1), spanning 2007–2025. Additionally, between September 2023 and January 2025, an RBR Concerto³ CTD-O was deployed at a depth of 5 m on the fjord shoreline, downstream of the Puelo River mouth (Fig. 1B, CTD), to assess the influ-

Variable	Source	Period	Frequency	Spatial coverage
Phytoplankton species and cell concentration	PROMOFI	1990–2024	Daily during HAB events	Northwestern Patagonia
Salmon mortality	SERNAPESCA	2012–2024	Weekly reports	Northwestern Patagonia
Climate (precipitation and air temperature)	ERA5 reanalysis	1960–2024	Daily	Puelo River Basin
Streamflow	DGA	1950–2024	Daily	Puelo River Basin
Water temperature	DGA and PME-MiniDOT sensor	2007–2025	10 min, hourly	Puelo River and Tagua Tagua Lake
Turbidity	Cyclops-7 sensor	Dec 2022–Jan 2025	10 min	Puelo River Basin (see Fig. 1)
Total Suspended Matter (TSM)	Sentinel-2	2016–2024	~ 5 days (image availability)	Tagua Tagua Lake
dSi, dFe, TN, TP, NH ₄ ⁺ , NO ₃ ⁻ , NO ₂ ⁻ , PO ₄ ³⁻ , TSS	FSEQ210030; FONDECYT 1,221,102	Apr 2022–Jun 2023; Oct–Dec 2024	Weekly, event-based	Puelo River Basin (see Fig. 1)
Water temperature, salinity, and dissolved oxygen	RBR Concerto ³ CTD-O	Sep 2023–Jan 2025	5 min	Reloncaví Fjord

Table 1. Summary of datasets used to evaluate the link between the timing and magnitude of meltwater-driven inputs from Puelo River and the development of HABs in the Reloncaví Fjord.

- ence of freshwater inputs on the surface layer. The instrument recorded temperature, salinity, and dissolved oxygen at 5-minute intervals.
- iii) Water-sampling campaigns: To resolve the biogeochemical composition associated with these hydrological and physicochemical signals, discrete water-sampling campaigns were conducted ~7 km downstream of the Carrera Basilio gauging station. These combined (a) a citizen-science monitoring program with approximately weekly sampling complemented by monthly validation by the research team (April 2022–June 2023), and (b) a high-frequency spring campaign (October–December 2024) designed to capture short-lived snowmelt-driven pulse events. The water samples were collected using a tethered bucket in the middle stream section with single and triplicate samples for citizen and project teams, respectively. The fieldwork in both initiatives involved collecting water samples for the analysis of Total Suspended Solids (TSS), dSi (orthosilicic acid), dFe, and nitrogen and phosphorus (N-total, P-total, NH_4^+ , NO_3^- , NO_2^- , and PO_4^{3-}). One-liter grab samples were collected with a bucket for syringe filtration *in situ*. The samples for dSi and dissolved dFe were filtered through a 60 mL syringe (0.45 μ m nitrocellulose filters). TSS was estimated by filtering 1 L grab samples through pre-weighed 0.7 μ m Whatman GF/F filters, followed by drying at 40 °C to constant weight and determination of mass change. Water samples were kept refrigerated or frozen (total and dissolved nitrogen and phosphorus nutrients) until analysis at the CIEP laboratory in Coyhaique. NH_4^+ was analyzed by ion chromatography (Dionex ICS 5000), together with colorimetric methods for NO_3^- following cadmium reduction, PO_4^{3-} by ascorbic acid method, TN/TP by persulfate digestion and colorimetric methods (APHA, 2005), dSi by modified molybdate method with Metol-sulfide reduction⁴⁷, and Fe (total and dissolved) by Atomic Absorption Spectrophotometry with Graphite Furnace (Perkin Elmer PinAAcle 900AH⁴⁷). The Si* index was calculated as the molar excess of dSi over inorganic nitrogen (presuming NOx is the major form of inorganic N; Torres et al.⁴⁸). Since dSi is almost universally well in excess for freshwater systems in Patagonia, its useful as a marine index depends on mixing ratios (see below), with potential Si* thresholds of 1.0 favoring diatom production, and 10 associated with diatom bloom conditions⁴⁹. Redfield proportions of riverine inorganic macronutrients were calculated as the molar ratio of inorganic nitrogen ($NOx + NH_4^+$) to soluble reactive phosphorus (PO_4^{3-}).
 - iv) Satellite-derived data: To extend the TSS record (2016–2024), Sentinel-2 Level-1 C (S2-L1C) imagery was retrieved from the Copernicus Programme. Total Suspended Matter (TSM) was extracted at three representative locations in Tagua Tagua Lake (Fig. 1, TTL): near the Puelo River inflow, the mid-lake zone, and the lake outflow. In this study, TSM was used as a proxy for TSS, acknowledging that it represents the optically active fraction of suspended particulate matter rather than a strictly equivalent measure. The S2 imagery was atmospherically corrected using ACOLITE (version 20231014) with the Dark Spectrum Fitting (DSF) method⁵⁰, complemented by shortwave infrared (SWIR)-based sunglint correction⁵¹. TSM concentrations were computed within ACOLITE using the implementation of the algorithm by Nechad et al.,⁵² applied to the Sentinel-2 red band ($\lambda = 665$ nm; Band 4, 10 m spatial resolution). For each sampling point, values were derived as the mean of a 3 × 3-pixel window centered on the target location. To ensure consistency across observation, a site-specific relationship between turbidity (sensor output in mV) and TSS (mg L⁻¹) was established using observations from the citizen-science monitoring program. This empirical calibration enabled the conversion of the continuous turbidity record into TSS-equivalent values, extending the temporal resolution of suspended sediment dynamics. During the overlapping period, turbidity-derived TSS and satellite-derived TSM exhibited similar magnitudes and temporal variability (see Fig. 8), suggesting that they can be used together to characterize suspended matter dynamics across different scales.

Statistical analysis

To analyze the seasonal and interannual variability of freshwater input, we updated previously computed statistics of the daily flow data of Puelo River using the R package *fasstr*⁵³, which computes a wide range of flow statistics, including daily, monthly, and annual percentiles; low- and high-flow extremes over different durations (1, 3, 7, 30 days); frequency and duration of exceedance events; and the timing of annual minima and maxima. To assess changes in hydrological indicators, temporal trends were quantified using Sen's slope⁵⁴ and tested significance with the Mann–Kendall test.

Cold events were identified using in-water sensor measurements and water temperature records (2016–2025) and were defined as periods of three or more consecutive days with daily temperature anomalies of at least –0.5 °C. Given the limited length of the river water-temperature record, we adopted a conservative moving-window approach^{55,56} based on a centered window of 22 days preceding and following each observation (45-day window total). To evaluate differences in physicochemical and flow variables, we applied nonparametric Mann–Whitney (Wilcoxon rank-sum) tests to data groups defined by monitoring periods and seasons.

To explore water quality–discharge (cQ) relationships, we analyzed log–log linear regressions between observed daily streamflow (Q) and physicochemical concentrations (c), following the framework proposed by Moatar et al.⁵⁷. Concentration and discharge data were log₁₀-transformed and fitted using ordinary least squares regression according to the model $\log_{10}(C) = \log_{10}(a) + b \cdot \log_{10}(Q)$, which corresponds to a power-law relationship of the form $C = a \cdot Q_{50}^b$. To account for potential changes in solute behavior across the flow regime, the dataset was segmented at the median discharge (Q_{50}), and the same regression framework was applied separately to data below ($b_{50,inf}$) and above ($b_{50,sup}$) Q_{50} . Regression slopes (b), coefficients of determination (R^2), and significance levels were extracted for each segment, with statistical significance assessed using Student's t-tests on the slope parameters.

Estimating the spatial distribution of riverine export and its potential effects on fjord productivity and HABs is challenging, as it is strongly controlled by variable river flows (spanning orders of magnitude), fjord hydrography, density gradients, tides, and wind. A solution normally requires complex, three-dimensional hydrodynamic modeling. In this context, we developed a simplified index to represent the potential spatial

influence of river export and mixing within the fjord, defined here as a mixing radius (R_m , km), representing the linear distance from the river mouth of a pre-defined mixing proportion (P , unitless), effective mixing depth (d , km), and assuming a near-zero marine end-member. Under idealized mixing conditions, R_m is proportional to $V_m^{1/2}$. The mixing radius was derived from monthly cumulative freshwater runoff volumes (V_m) for the 2023, 2024, and 2025 water years, and assumes that freshwater is distributed within a semi-cylindrical surface layer extending offshore from the river mouth. We estimated R_m assuming d (50 m) and a range of P (0.8, 0.9 and 0.95).

Results

Baseline hydrological conditions and long-term trends

Long-term analysis of the Puelo River daily flow series (1950–2024) reveals a consistent drying trend across multiple hydrological indicators (Table S1). Total annual discharge, mean flow, and runoff volume declined significantly during austral summer and autumn (December–March and May; Fig. 2A–C), alongside a reduction in the number of normal-flow days ($\pm 20\%$ of annual mean; Fig. 2D). Low-flow metrics also indicate progressively reduced flow conditions (Fig. 2E–F; Table S1). The Julian day of the annual 30-day maximum flow shifted significantly from winter toward early spring (Fig. 2F), consistent with earlier snowmelt timing driven by regional warming.

Puelo River temperature at the downstream station (Fig. 1) showed a coherent seasonal cycle over the 2007–2025 period, with minima in August–September and maxima in January–February (Fig. 3D). Throughout 2016–2025, 23 cold-anomaly events were detected (median: 7 days; median anomaly: -0.7°C , Table S2), generally in November and January. Along the basin, the temperature at the upstream stations (PR1 and MR) responded rapidly to storm pulses, whereas Tagua Tagua Lake dampened and phase-lagged these thermal signals (Fig. S1).

Physicochemical signature of Puelo River freshwater export

Concentration–discharge (cQ) analysis over 2022–2024 revealed a positive response of trace nutrients (dFe) excess silica (Si^*) and stoichiometric macronutrient ratios (N:P inorg.) to river flow (Fig. 4). Dissolved iron (dFe; $\mu = 0.33 \mu\text{mol L}^{-1}$) exhibited a weak, non-significant positive association with flow, whereas excess silica (Si^* ; $\mu = 104.1 \mu\text{mol L}^{-1}$) showed a moderate increase with flow that was significant (Fig. 4). Inorganic molar

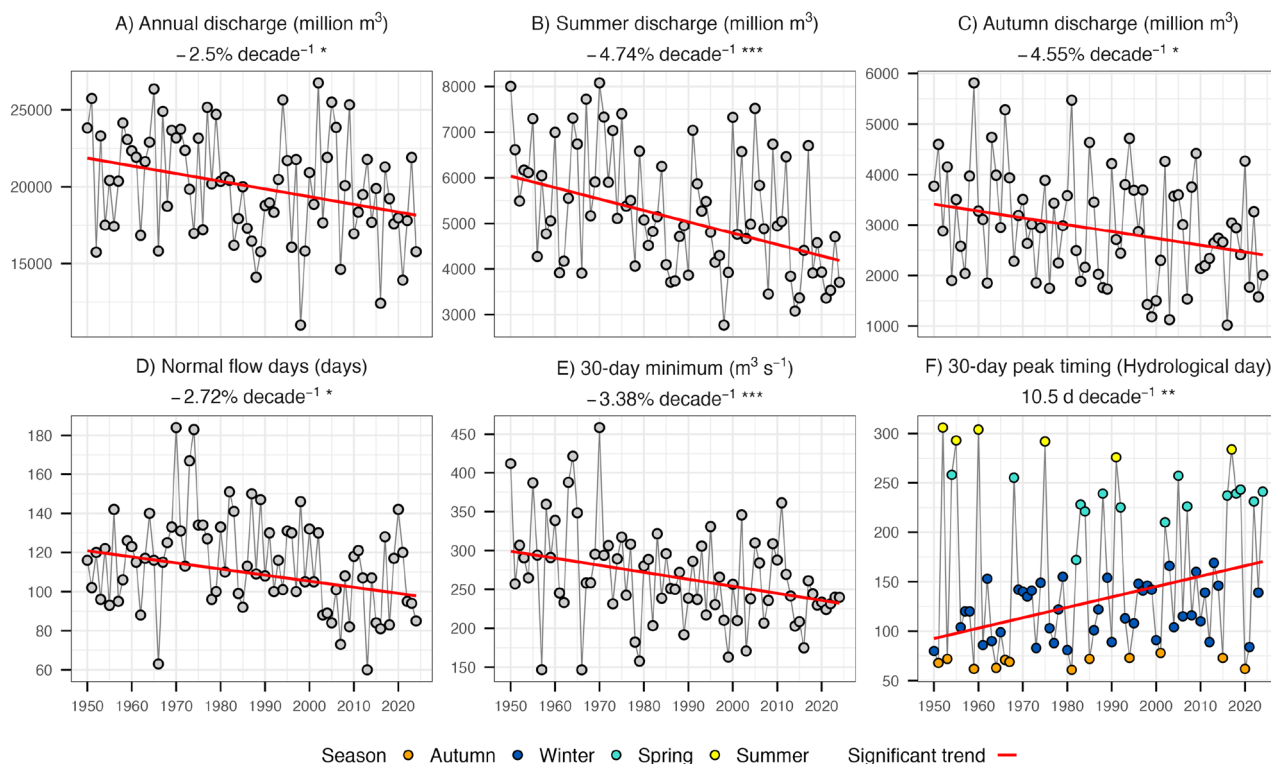


Fig. 2. Long-term significant trends (red lines) in Puelo River hydrological indicators (water years 1950–2024), including volumetric discharge (million m^3) at annual (A), summer (B), and autumn (C) scales; the number of normal-flow days (D); daily discharge within the central percentile range, e.g., Q10–Q90; the 30-day minimum flow (E); and the 30-day peak timing, defined as the day of the year (Apr 1 = 0) when the maximum 30-day moving-average discharge occurs (F). Statistical significance is represented by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Colored circles in the 30-day peak-flow timing panel represent the season in which the annual maximum occurs. Detailed values of slope, intercept, and statistical significance are provided in Table S1.

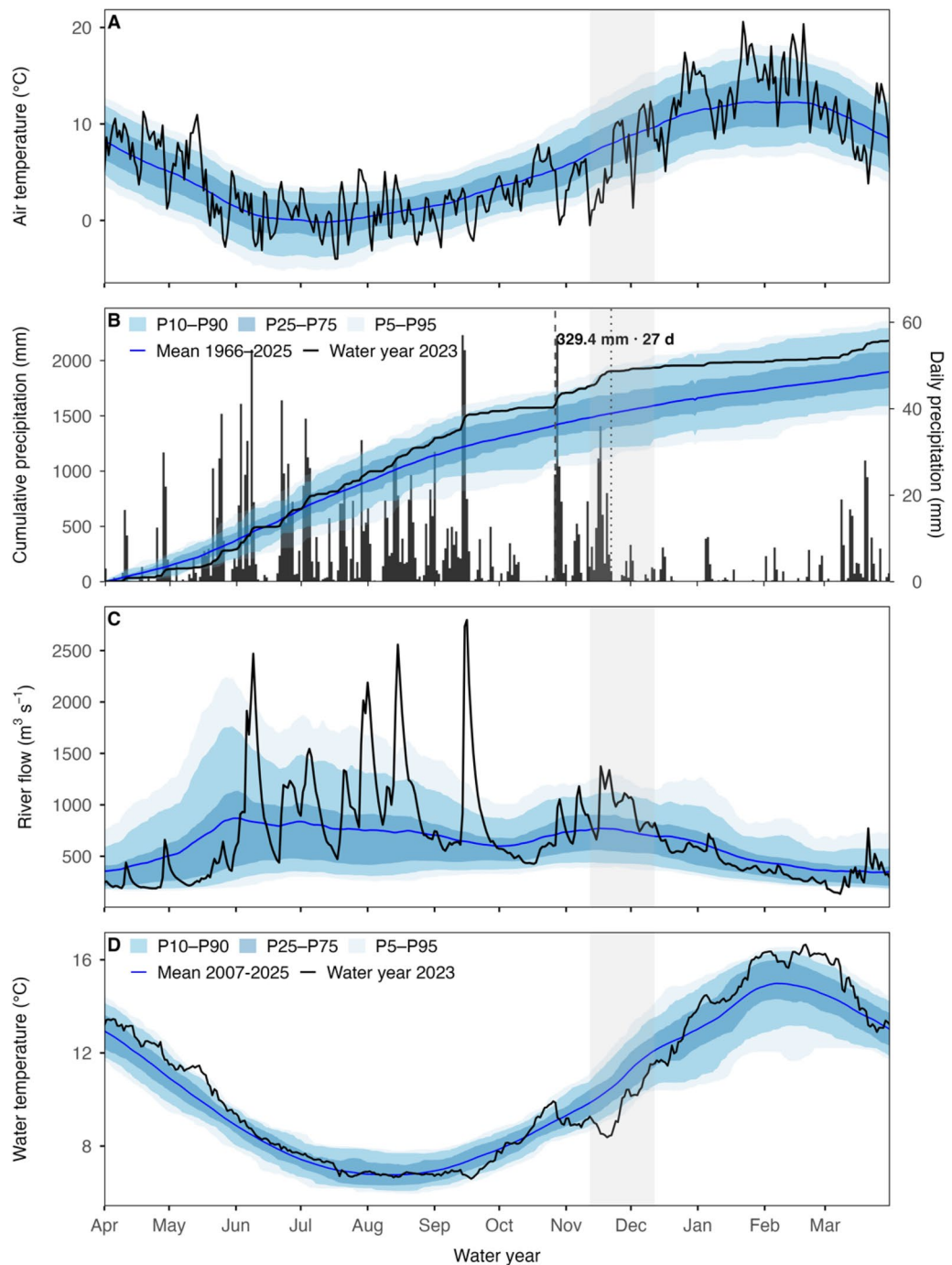


Fig. 3. Climatology of hydro-meteorological variables for the 2023 water year, in relation to historical data (1966–2025). Panels (A) and (B) illustrate the daily air temperature and precipitation data obtained by study area from the ERA5 reanalysis. Panels (C) and (D) show daily streamflow (1966–2025) and water temperature (2007–2025) data collected at the Carrera Basilio gauging station (Fig. 1). The shaded envelopes denote the percentile ranges, and the blue line represents the long-term mean for each variable while the black line corresponds to the 2023 water year. The vertical gray band marks the November 2023 HAB event. In Panel (B), the solid and dotted vertical lines indicate the start and end of the major precipitation event recorded between late October and November 2023.

N:P ratio ($\mu = 14.9$), showed a strong and significant response to flow (Fig. 4), a combined response of inorganic nitrogen ($\text{NO}_x + \text{NH}_4^+$) which increased significantly with discharge (slope = +0.551; $R^2 = 0.13$, $p < 0.05$), versus inorganic phosphorus (PO_4^{3-}) with a negative association (slope = -0.209; $R^2 = 0.07$) (Fig. S2). Total Suspended Solids (TSS; $\mu = 9 \text{ mg L}^{-1}$) estimated from in situ monitoring and satellite-derived data also exhibited a significant positive response to flow, while the model was improved with split regression, with a threshold near $500 \text{ m}^3 \text{ s}^{-1}$

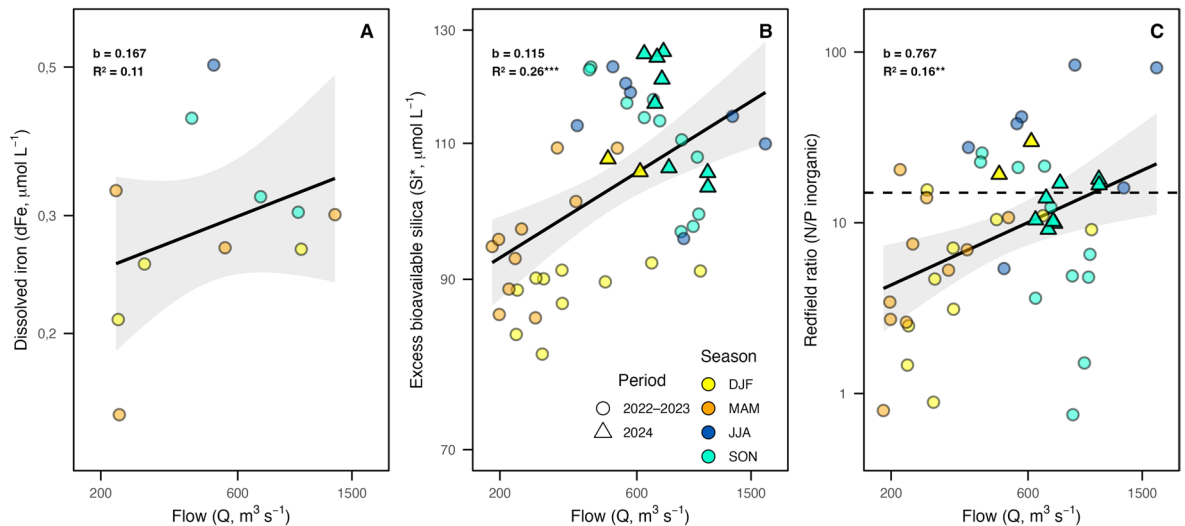


Fig. 4. cQ relationships between mean daily flow (Q) and nutrient/resource concentrations of river export potentially linked to marine primary production: dissolved iron (dFe), excess dissolved silica (Si^*) and Redfield ratio (16:1, $NO_x + NH_4^+ : PO_4^{3-}$). Colors indicate the season of each sampling date, while circles represent baseline conditions (2022–2023) and triangles highlight samples associated with the spring 2024 pulse. Solid lines show the fitted log–log ordinary least squares (OLS) regressions, with shaded areas representing 95% confidence intervals. Regression slopes (b), coefficients of determination (R^2), and significance levels ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$) are indicated in each panel. All concentrations are in $\mu\text{mol L}^{-1}$, while N:P ratio of 16:1 shown as dashed line.

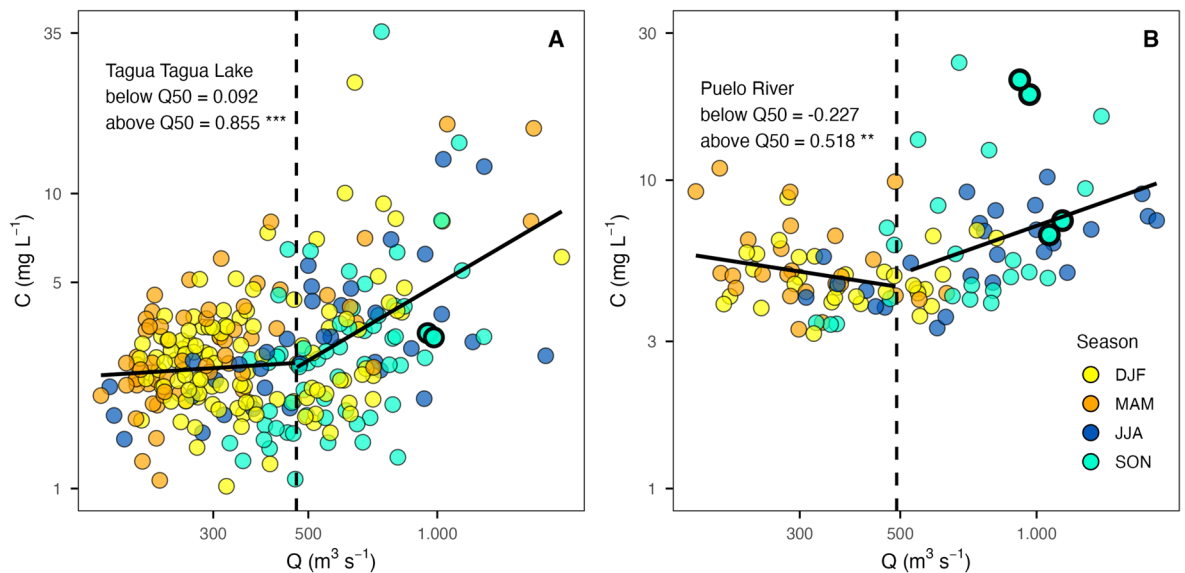


Fig. 5. cQ relationships between daily discharge (Q , $\text{m}^3 \text{s}^{-1}$) and Total Suspended Solids (TSS) concentrations (C , mg L^{-1}) for Tagua Tagua Lake (A, 2016–2024) and the Puelo River (B, 2022–2024). Colors represent season for each sampling date, while bold outlines represent November 2023 pre-bloom conditions. Slopes (b) for split regressions, separated based on median sample flow (dashed line), and significance levels ($***p < 0.001$; $**p < 0.01$; $*p < 0.05$) are indicated in each panel. p -values correspond to the t -test on the regression slope from the OLS fit.

flows for both stations (Fig. 5). Elevated TSS values and positive cQ responses were concentrated predominantly in the austral spring period.

Seasonal patterns in river runoff water quality from the two sampling time series bracketing the 2023 event indicate the likely range of inorganic nutrient contributions from the Puelo River to Reloncavi Fjord (Fig. 6). During the first observation period (August 2022–July 2023), Si^* concentrations were highest in winter and early spring (July–October 2022; $>110 \mu\text{mol L}^{-1}$) and lowest in mid-summer (January–February 2023; $\sim 80 \mu\text{mol}$

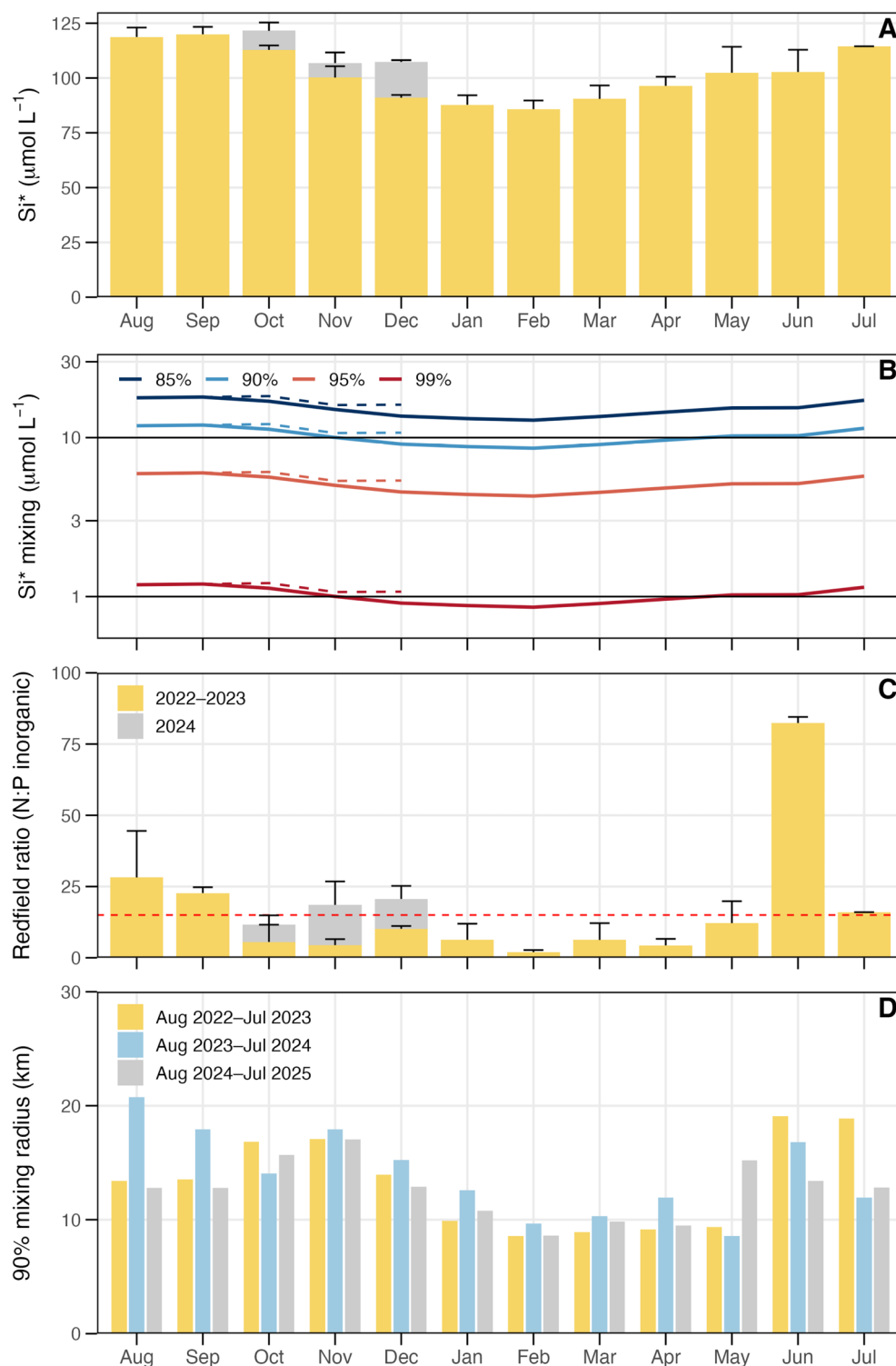


Fig. 6. Resource ratios and the potential influence of river–marine mixing across the annual cycle in the Puelo River system: **(A)** Si* for river input, defined as the excess of dissolved silica (dSi) relative to inorganic nitrogen. **(B)** Hypothetical Si* concentrations resulting from progressive mixing with Reloncaví Fjord marine water between 2003 and 2022 (solid lines) and the 2024 sampling period (dotted lines). Potential Si* thresholds shown for 1.0 favoring diatom production, and 10.0 potentially associated with diatom bloom conditions. **(C)** Monthly molar ratio of inorganic nitrogen (NO_x + NH₄⁺) to soluble reactive phosphorus (PO₄³⁻). The Redfield ratio (16:1), shown as a red dashed line, provides a reference threshold for identifying potential nitrogen or phosphorus limitation based on river inputs. **(D)** Potential spatial influence of river export mixing within the fjord, in this example estimated for the 90% mixing of seawater with 10% river water and a 50 m mixing depth.

L^{-1} (Fig. 6A), while during spring 2024, Si^* values were approximately 5–10% higher than during the same months in 2022. Theoretical mixing scenarios (85–99% dilution) indicate that, under strong dilution (99%), the annual cycle of riverine Si^* remains slightly below the $1 \mu\text{mol L}^{-1}$ threshold, except in late winter and early spring 2022, whereas it remains above that threshold throughout the corresponding period in 2024. At 90% dilution, a similar seasonal pattern is observed, with Si^* values fluctuating around the bloom-related threshold of $10 \mu\text{mol L}^{-1}$ (Fig. 6B). Over 2022–2023, the inorganic N:P ratio frequently fell below the Redfield proportion (16:1) during spring, summer, and much of autumn 2022, whereas the highest values, indicating excess inorganic N, occurred in winter ($>60:1$) and early spring ($\sim 25:1$) (Fig. 6C). In contrast, during the 2024 spring snowmelt, N:P ratios were consistently and significantly higher than in 2022, remaining above 16:1 through November and early December (Fig. 6C). The modeled 90% mixing radius for Si^* also varied seasonally and between years (Fig. 6D), with winter 2023 showing the broadest area of influence relative to 2022 and 2024, followed by a marked contraction in October 2023 and a rebound in November. Because this scenario corresponds to an Si^* threshold near $10 \mu\text{mol L}^{-1}$, it provides an estimate of the potential spatial extent of this condition within the fjord relative to river input; by contrast, riverine N:P, expressed as a ratio, would remain unchanged under simple dilution assumptions.

At finer temporal resolution, winter–spring conditions in 2022 and 2024 reveal contrasting hydroclimatic and nutrient dynamics (Fig. 7). In 2022, air temperature anomalies were neutral to slightly positive, while river temperature anomalies were slightly negative. During this period, inorganic N:P ratios remained below 16:1, and Si^* progressively declined from late winter, dropping below $10 \mu\text{mol L}^{-1}$ from mid-November onward. By comparison, spring 2024 was characterized by negative air temperature anomalies, including a cold spell from mid-November to early December (Fig. 7). Throughout this interval, Si^* remained elevated above $10 \mu\text{mol L}^{-1}$, while inorganic N:P ratios showed short-lived peaks exceeding 16:1. More broadly, the October–November 2024

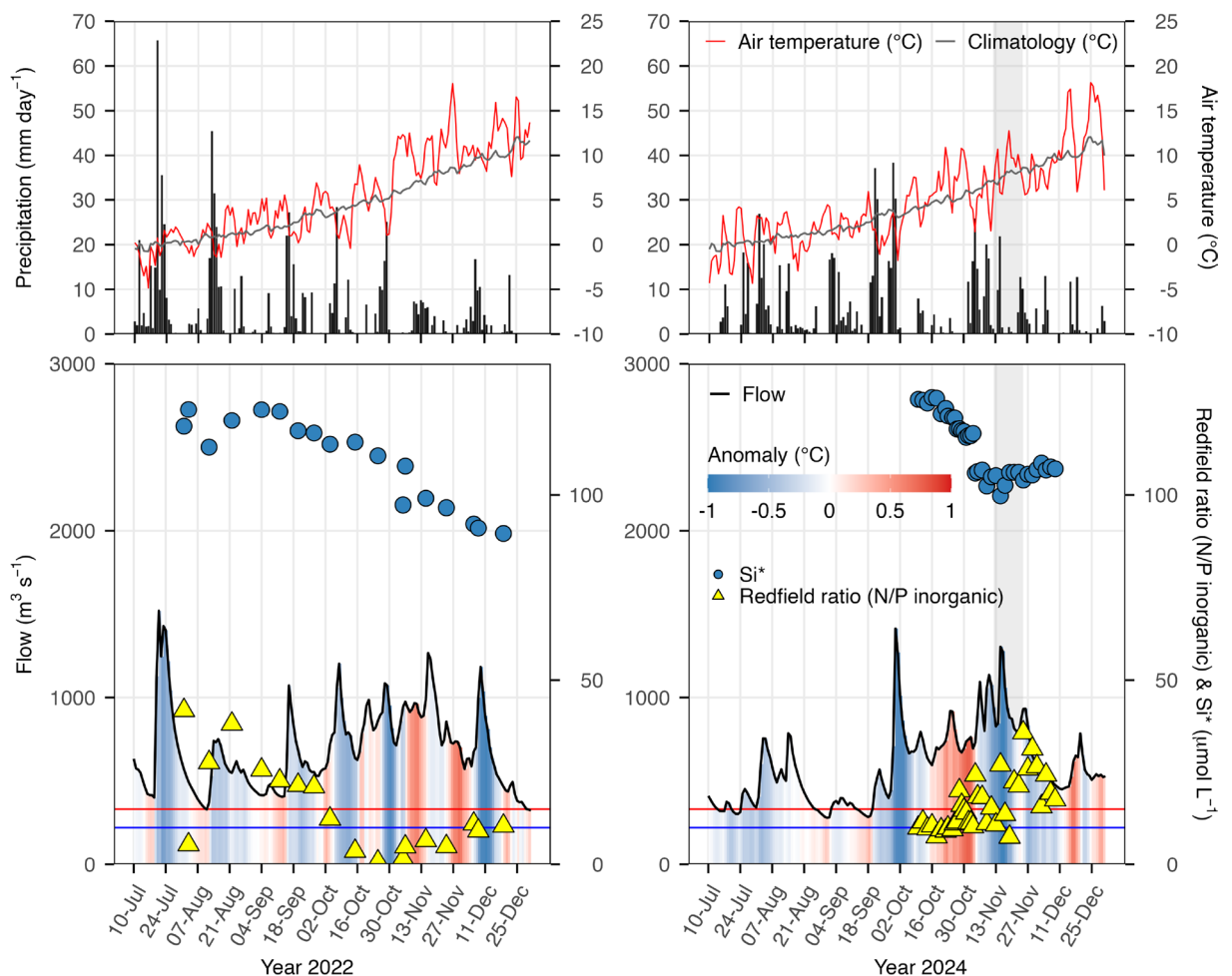


Fig. 7. Excess bioavailable silica (Si^* , blue circles) and the inorganic Redfield ratio (N:P = 16:1, yellow triangles) during the winter–spring period of water years 2022 and 2024. Upper panels display daily precipitation (bars) and air temperature obtained from the ERA5 reanalysis, with the black line indicating the long-term daily mean (1960–2015). Lower panels show Puelo River flow (black line) and river water temperature anomalies ($^{\circ}\text{C}$). The red horizontal line denotes the Redfield threshold (N:P = 16:1), while the blue horizontal line marks the Si^* threshold (10), associated with favorable conditions for HAB development. Vertical gray shading highlights a period of elevated spring abundance of *Thalassiosira cf. pseudonana* in 2024.

sampling period captured two successive snowmelt-driven runoff events, marked by rapid shifts in nutrient ratios, increased discharge associated with precipitation, and negative river-temperature anomalies.

Anomalous hydro-meteorological and freshwater forcing in 2023

The 2023 water year represented a marked departure from the surrounding dry years and from long-term climatology (Fig. S3). Annual precipitation exceeded the long-term mean by ~34% (Fig. 3B), driven by exceptional winter and spring anomalies (+48% and +82%, respectively). In contrast, austral summer precipitation was markedly drier (–50% relative to both references), and autumn was only marginally above average (+12%). Annual mean discharge reached $693 \text{ m}^3 \text{ s}^{-1}$, approximately 10% above the long-term mean (Fig. 3C), with fewer below-normal days (95–120) than adjacent years. Winter–spring flood pulses (June–September) frequently exceeded the 95th flow percentile and coincided with above-average air temperatures (near historical P95), pointing to a rainfall-dominated hydrological regime (Figs. 3A, C). A sustained dry spell occurred between mid-September and mid-October, associated with a persistent blocking of the midlatitude flow in the southern part of the continent, that was broken by another precipitation period from the end of October and November (totaling 55 mm and 105 mm, respectively; Fig. 3B) causing flows above $1,000 \text{ m}^3 \text{ s}^{-1}$ (Fig. 3C). This pulse developed during a prolonged period of below-average air temperatures (Fig. A), which likely prolonged both the melting process and the snowmelt flow pulse (Fig. 3C, D). In terms of water temperature, two cold river-temperature anomaly events were recorded in 2023: one in September (7 days, -0.71°C) and a stronger one in November (11 days, -1.34°C). The November event was among the most pronounced cold anomalies in the 2016–2025 record (Fig. 3D; Table S2) and propagated coherently through all basin stations, including Tagua Tagua Lake; in contrast to most cold events, which are attenuated at the lake (Fig. S1).

Nutrient dynamics during this period were similarly anomalous. Inorganic N:P ratios in spring 2024, which shared the closest hydrological analogy to the pre-bloom 2023 conditions, remained consistently above the Redfield threshold through November and into early December, with short-duration peaks substantially exceeding 16:1 (Figs. 6C and 7). TSS concentrations in the pre-bloom window were among the highest observed across the full study period, placing the November 2023 freshwater inputs in the upper tail of the cQ distribution (Fig. 5, bold outlines). Taken together, the convergence of elevated TSS, above-Redfield N:P, sustained Si^* above the bloom threshold, and a pronounced cold snowmelt river-temperature signature defines a distinct physical and chemical template for the November 2023 period.

HAB dynamics and their relationship to freshwater forcing

Nearly 50,000 tons of salmon mortality over the last decade (SERNAPESCA, 2012–2024) correspond with HAB events (PROMOFI, 1990–2024), involving three principal taxa: the dictyochophyte *Pseudochattonella verruculosa* in 2016 (February–March >40,000 tons), the raphidophyte *Heterosigma akashiwo* in 2021 (March–April >6,000 tons), and the diatom *Thalassiosira* cf. *pseudonana* in 2023 (November ~1,500 tons). First documented as a potential HAB in 2016 in the Aysén Fjord⁴¹ (45°S), *T. cf. pseudonana* has since been repeatedly associated with fish mortality in the Reloncaví Fjord, with events increasing in frequency (Fig. S4). Its genetic identity in Chilean waters was not formally established until Paredes-Mella et al.⁴² isolated strain TP.Rel01 from the November 2023 Reloncaví bloom and confirmed species assignment through large-subunit rDNA phylogenetics and scanning electron microscopy. In relation with freshwater inputs from the Puelo River in the Reloncaví Fjord, *P. verruculosa* HABs occurred predominantly during late-summer to early-autumn low-flow conditions, whereas *T. cf. pseudonana* HABs spanned austral spring, summer, and autumn and developed across a broader range of flow regimes (Fig. 8).

Discussion

Not everything happens in drought scenarios

In NWP the most severe HAB events impacting aquaculture, including the 2016 *Pseudochattonella verruculosa* and 2021 *Heterosigma akashiwo* events, have consistently been associated with prolonged dry periods, increased solar radiation, weakened surface stratification and supply of inorganic nutrients from subsurface seawater^{58,59}. The results presented here indicate that additional mechanisms such as episodic high-flow pulses may also trigger bloom conditions. These findings do not imply that high-flow pulses alone are responsible for HAB occurrence, but rather that spring melt-driven pulses that were historically characteristic of the natural flow regime may, under increasingly dry conditions, emerge as key precursors of HAB development. Specifically, the conditions leading up to the November 2023 *T. cf. pseudonana* HAB event, associated with approximately 1,500 tons of salmonid mortality (>US\$10 million at average export prices), can be interpreted as the result of four interacting mechanisms linking freshwater inputs to changes in stratification, nutrient availability, and ultimately HAB development:

- 1) The sustained elevated freshwater discharge observed during winter 2023 (June–September; Figs. 3B and 6D) is consistent with enhanced halocline–pycnocline stratification^{19,60} and increased exchange flow at the Reloncaví fjord–sound interface⁶¹, potentially influencing nutrient transport and availability.
- 2) Cooler-than-average conditions from mid-September to mid-October (Fig. 3A) likely delayed basin-wide snowmelt, contributing to a temporal shift in the spring meltwater-driven pulse (Fig. 3C).
- 3) While stratification can limit the vertical mixing of nutrient-rich marine waters, it may be intermittently overcome by wind, tides, or, as argued here, high-discharge riverine pulses. Intense rainfall in late October (Fig. 3A, B) abruptly ended a preceding period of dry, warm, and high-radiation conditions, triggering short-lived freshwater pulses that likely modified stratification, light penetration, and nutrient availability. Together, these conditions likely favored the development of early blooms or pre-bloom community structure.

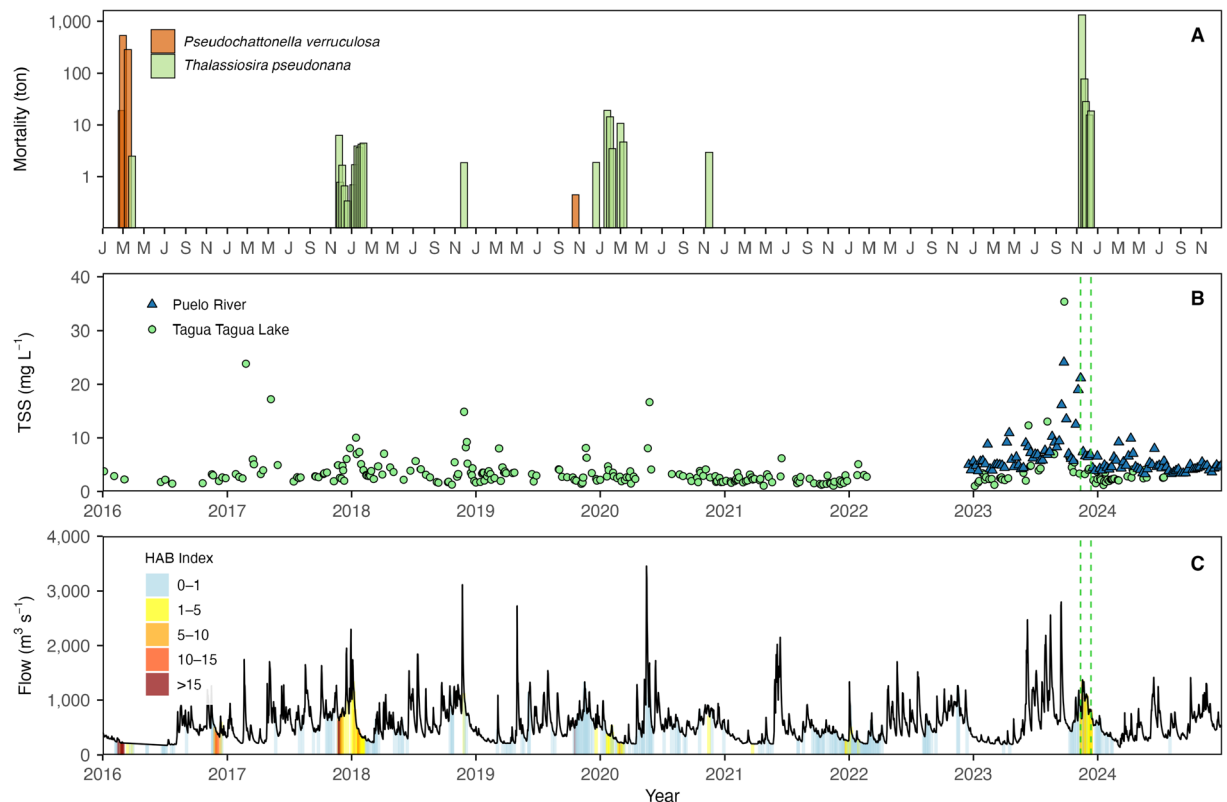


Fig. 8. Seasonal dynamics of Puelo River's flow and harmful algal bloom (HAB) records in the Reloncaví Fjord. **(A)** Temporal distribution of salmon mortality events (tons, log scale) attributed to *Pseudochattonella verruculosa* and *Thalassiosira cf. pseudonana*. **(B)** Total Suspended Solids (TSS) derived from Sentinel-2 observations in Lake Tagua Tagua (green circles; Fig. 1B) and from in-situ turbidity sensors in the Puelo River downstream of the lake (blue triangles). **(C)** Daily flow of Puelo River (black line), the colors beneath the discharge curve represent the HAB index. The dashed vertical green lines highlight the 2023 HAB event.

- 4) Intense rainfall in early November surpassed the retention capacity of Tagua-Tagua Lake (Fig. S1) and triggered an 11-day cold, dense discharge pulse (minimum -1.34°C ; mean -1.0°C ; Figs. 3 and 9). This pulse was associated with a rapid downstream propagation of cold freshwater (Fig. S1), which transiently altered the structure and persistence of the surface layer (Fig. 9). This cold-pulse was accompanied by elevated turbidity (Fig. 5), and, based on observations from comparable conditions in 2022–2023 and 2024, likely with excess silica (i.e. Si^* , Fig. 4) and favorable macronutrient ratios.

These processes culminated in a biological response over the following 1–2 weeks, with the initial phase of the *Thalassiosira cf. pseudonana* HAB coinciding with the high-turbidity pulses, and peak salmon mortality occurring during the coldest phase of the river anomaly (Fig. 9). This sequence supports a process-based framework in which HAB initiation is better explained by the combined effects of physical forcing and nutrient dynamics than by a single driver. In this context, the meltwater-driven pulse itself may not have been exceptional; rather, its occurrence following a sequence of antecedent hydrometeorological conditions that are becoming increasingly rare likely created a transient window of favorable conditions that enabled the HAB event in the fjord.

Citizen-science monitoring in the Puelo River indicates that high meltwater-driven pulses are consistently associated with moderate-to-high dFe concentrations, Si^* enrichment, inorganic N:P ratios exceeding 16:1, and reduced light availability driven by elevated turbidity (see c–Q relationships Figs. 4 and 5). These observations provide a process-based basis to infer the hydro-biogeochemical conditions associated with meltwater-driven pulses during the November 2023 event, despite the lack of direct water-quality measurements for that period.

Collectively, these short-lived windows of nutrient availability are consistent with a selective advantage for fast-growing, r-strategist diatoms over other phytoplankton groups. Cold, silica-rich pulses can reset surface-layer conditions by lowering temperature, modifying stratification, and enhancing dSi availability an environmental combination that typically favors opportunistic diatoms over slower-growing flagellates. Because *Thalassiosira* species are capable of rapid “luxury uptake” (supracellular quota, $Q > Q_0$) of silicic acid and can tolerate low-light conditions associated with high turbidity, these transient nutrient and optical shifts likely gave this taxon a short competitive advantage over other diatoms present in spring assemblages^{62,63}. Recent omics-based work with *T. cf. pseudonana*⁶⁴ also shows that, under low irradiance, this diatom can redirect carbon and energy fluxes toward pathways that conserve photosynthetic energy and enhance metabolic readiness, a strategy that may allow cells to persist through turbid, light-limited pulses and respond quickly when conditions improve.

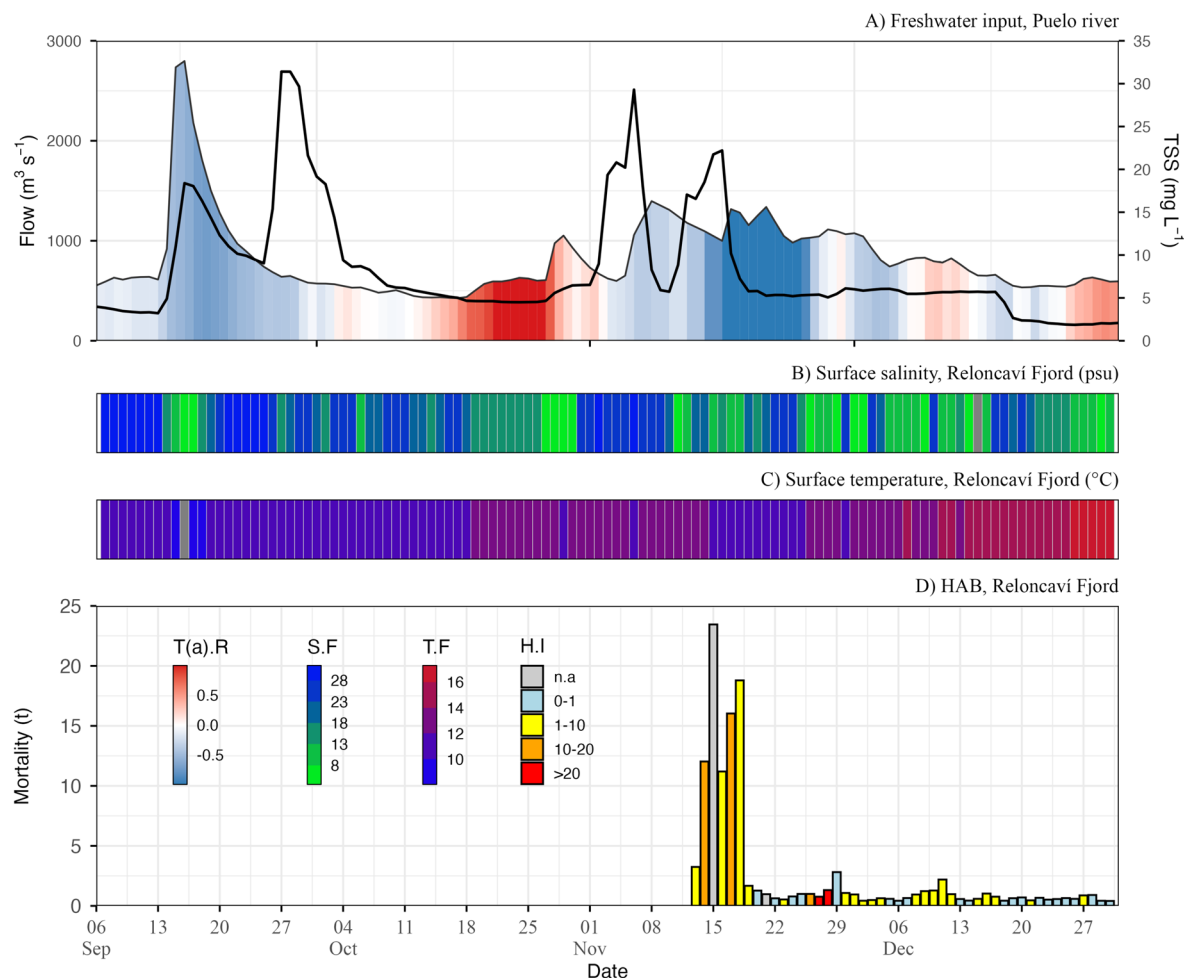


Fig. 9. Land-sea synchrony during the *Thalassiosira cf. pseudonana* HAB that occurred in the Reloncavi fjord in November 2023: **(A)** Total Suspended Solids (TSS, right axis) and daily flow records of Puelo River in its final section before entering the Reloncavi Fjord. The colored band below the flow-series represents the water temperature anomaly (T(a).R) relative to the 60-day moving average. **(B)** Surface salinity (S.F) and **(C)** temperature (T.F) measurements in the Reloncavi Fjord (5 m depth). Color scales for these figures are in Figure D below. Figure D also shows bars representing daily salmon farm biomass mortality (in tons) reported by one of the affected salmon farm facilities. The colors of the bars represent the HAB index (H.I), which is the ratio of the observed cell concentrations to the HAB lethal concentrations ($3,000 \text{ cells L}^{-1}$).

Together, these observations support a process-based framework in which meltwater-driven freshwater inputs act as both physical and biogeochemical drivers of HAB in the Reloncavi Fjord. At the seasonal scale, sustained winter runoff preceded by dry conditions influence stratification and surface-layer stability. At the short-term event scale, late spring discharge pulses modify thermal and salinity structure of the surface layer, increase suspended sediment export, and alter nutrient delivery and stoichiometry. The interaction between these processes creates transient environmental conditions favorable for a diatom bloom initiation and intensification.

Process-oriented assessments that explicitly couple hydrological forcing, nutrient dynamics, and HAB development remain scarce in Chilean fjords^{31,65}. One of the few examples is the 2021 *Heterosigma akashiwo* bloom in Comau Fjord, where unusually low N:P ratios, the preferential use of reduced inorganic and organic nitrogen forms, and a mixotrophic strategy allowed a persistent, high-biomass flagellate bloom under overall low inorganic nutrient conditions⁶⁶. When viewed together, the Puelo-Reloncavi case, in which flood-driven inputs of dSi, dFe and terrestrial DOM favor diatom-dominated HABs, and the Comau *H. akashiwo* event suggest that subtle differences in nutrient stoichiometry, nitrogen speciation, and carbon availability, superimposed on river-driven stratification, can steer fjord systems toward contrasting HAB trajectories (diatom vs. flagellate dominated).

In this context, although diatoms are usually known as strictly phototrophic, several species have recently shown mixotrophic or metabolically flexible behaviors under certain conditions, including the ability to use external organic carbon sources^{67,68}. This added trophic versatility, documented for genera such as *Phaeodactylum*, may further modulate their response to freshwater pulses enriched in DOM, highlighting that metabolic plasticity is not restricted to flagellates. Taken together, these insights underscore the need for more

process-level studies that integrate hydrological forcing, nutrient dynamics, and phytoplankton functional traits when evaluating HAB risk in Patagonian fjords.

Potential mechanisms underlying fish mortality in *T. cf. pseudonana* HAB

Although *T. cf. pseudonana* is not recognized as a toxin-producing species, its capacity to form dense aggregations and secrete extracellular polymeric substances (EPS) under sudden environmental stress could have contributed to gill obstruction in farmed fish, particularly when combined with elevated TSS and abrupt temperature declines that increase respiration demand⁶⁹. At present, the specific mechanism by which *T. cf. pseudonana* contributes to fish mortality remains unresolved. Unlike several ichthyotoxic flagellates, where mucus production, reactive oxygen species, or direct cell-gill interactions have been documented, the pathways leading to fish kills in diatom-dominated blooms are poorly characterized. Most known ichthyotoxins differ sharply from classical phycotoxins affecting human health, and their chemical identity remains largely unknown⁷⁰. Importantly, *T. cf. pseudonana* does not possess obvious physical traits to generate mechanical gill damage, such as the heavily silicified spines or barbs seen in taxa like *Chaetoceros concavicornis* or *Chaetoceros convolutus*, although field and culture observations from the 2023 event showed that cells aggregated and released mucus-like substances at high abundances⁴². In contrast, the production of EPS has been widely implicated in fish-killing episodes caused by raphidophytes and dinoflagellates, including *Heterosigma akashiwo*⁷¹, *Akashiwo sanguinea*, *Karenia mikimotoi*, and *Chattonella marina*^{69,72}, where mucilaginous exudates accumulate around gill lamellae and impair ventilation. Diatoms have rarely been associated with this mechanism, yet our observations, together with the extreme TSS and cold-pulse conditions, suggest that an EPS-mediated pathway cannot be dismissed. Paredes-Mella et al.⁴² tested this directly, screening a strain isolated from the same November 2023 event (TP. Rel01) for paralytic shellfish toxins, lipophilic toxins, and domoic acid (all negative) before exposing it to the RTgill-W1 rainbow trout gill cell line. Cell viability nonetheless declined significantly with abundance ($p < 0.01$), from $105.8 \pm 4.4\%$ at $8.6 \text{ cells mL}^{-1}$ to $82 \pm 2.8\%$ at $86,000 \text{ cells mL}^{-1}$, and cultures at high density aggregated visibly, releasing mucilaginous material. Such exudates raise water viscosity and restrict oxygen transfer across gill surfaces⁷³; the small diameter of *T. cf. pseudonana* cells sharpens this effect, as dense suspensions can occlude lamellar surfaces without producing classical ichthyotoxins.

Another potential explanation arises from recent hypotheses linking nutrient limitation with shifts in diatoms nutritional strategies. In Comau Fjord Mardones et al.⁵⁹ reported that nitrogen stress in *H. akashiwo* was accompanied by a rise in EPS, which drew in nearby bacteria and particles. Although this behavior is far less documented in diatoms, it offers a useful parallel for the conditions observed here. Although mixotrophy in diatoms is much less documented, emerging work highlights that several species exhibit metabolic flexibility and may use extracellular matrices to modulate interactions with heterotrophic bacteria^{68,74}. Such processes remain largely unexplored in *Thalassiosira*, but they offer a plausible mechanistic avenue: enhanced EPS secretion under nitrogen-poor, high-turbidity, low-light conditions may have intensified mucus loads in the water column, thereby contributing mechanically to gill fouling. Together, these considerations add to the knowledge gap regarding how small centric diatoms such as *T. cf. pseudonana* can impair gill function and trigger mortality. The mechanism is still uncertain, but EPS formed under nutrient stress or sharp environmental changes may have played a role and deserves a closer look.

Regional hydrological shifts and their implications for HAB Forcing in NWP

Although this study focuses on the Reloncaví Fjord–Puelo River system (Fig. 1), the mechanisms identified here are likely representative of broader land–sea interactions across NWP. Recent studies indicate that similar hydrological shifts are occurring in other major rivers of the region, including the Yelcho, Palena, Cisnes, and Aysén rivers¹⁹, suggesting a coherent regional signal.

From a biogeochemical perspective, rivers in NWP are characterized by high concentrations of dFe ($8.4\text{--}243.5 \text{ ug L}^{-1} \text{ Fe}$) and dSi ($0.5\text{--}12.6 \text{ mg L}^{-1} \text{ SiO}_2$), coupled with very low concentrations of TP ($2.8\text{--}523.7 \text{ ug L}^{-1}$) and TN ($6.4\text{--}705.6 \text{ ug L}^{-1}$), representing some of the lowest inorganic nutrient conditions reported globally⁷⁵. Such conditions reinforce the potential for riverine influence on coastal productivity and HAB dynamics, particularly considering the coastal dilution potential generated by these freshwater inputs.

However, large rivers are only part of the picture. León-Muñoz et al.¹⁹ estimate the average river's freshwater flux to NWP's coastal systems at $7,300 \text{ m}^3 \text{ s}^{-1}$, of which 52% originates in small- and medium-sized basins, while the remaining 48% is supplied by the six largest basins (Fig. 1). The smaller basins exhibit contrasting hydrological regimes, while large rivers are dominated by meltwater-driven spring–summer pulses, small coastal basins respond rapidly to precipitation, sustaining winter-dominated flows. This results in a bimodal freshwater delivery system at the land–sea interface.

The divergent hydrological responses among basin sizes, highlight a critical but underexplored dimension of HAB forcing in NWP fjords. Current approaches largely rely on discharge data from large rivers or, on average model outputs, thereby overlooking the contribution of smaller basins. The potential synergistic effects of meltwater pulses and rainfall-driven coastal inputs remain a key knowledge gap.

Despite these uncertainties, our results demonstrate that meltwater-driven freshwater inputs can alter the physical and biogeochemical structure of the surface layer of semi-enclosed coastal systems and create conditions conducive to HAB development. In particular, the anomalously high and persistent spring pulse observed before the November 2023 *Thalassiosira cf. pseudonana* HAB suggests that hydrological signals could forecast HAB onset by several days to over a week. This finding underscores the potential of hydrological monitoring as a basis for early-warning systems in aquaculture-relevant fjord environments. A temporal lag between basin-scale hydrological processes and their downstream expression in the fjord provides a measurable lead time, which, although limited, may be operationally significant, as it could enable the timely activation of mitigation strategies (e.g., intensified PROMOFI monitoring). Expanding monitoring networks upstream, including tributaries and

lake systems, could further enhance this capacity, as hydrological anomalies originate in headwater basins before propagating downstream.

Finally, our results indicate that HAB development in the Reloncaví Fjord arises from the coupled interaction of climatic and hydrological forcing, stratification dynamics, nutrient availability, and biological response, rather than from any single controlling factor. In the case of the 2023 *Thalassiosira cf. pseudonana* HAB, the critical factor was not the magnitude of an individual meltwater pulse, but its occurrence within a specific sequence of previous hydrometeorological conditions that created a transient window of favorable environmental conditions. As such sequences become increasingly infrequent under ongoing climatic change, their ecological consequences may become more pronounced. This highlights the need to move beyond event-based interpretations toward a process-based understanding of basin–fjord connectivity, with direct implications for forecasting and reducing HAB risk across NWP fjords. More broadly, our findings suggest that river–coastal coupling is an important driver of HAB dynamics, with potential relevance for understanding and anticipating HAB risk in fjords and estuaries at other latitudes.

Data availability

Observed streamflow time series were obtained from the General Directorate of Water (DGA; <https://dga.mop.gob.cl>). Meteorological forcing data were derived from the ERA5 reanalysis. The dataset of water quality used in this study is available at <https://zenodo.org/records/16422064>. Specific data may be requested from the corresponding author of this paper.

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Author contributions

J.L.-M. conceived the study and led the manuscript writing. B.R. contributed to manuscript development and conceptual refinement. J.I., J.M., D.S., R.A., A.A., and R.G. contributed to the analysis and interpretation of hydro-meteorological and oceanographic data. W.G.-T. contributed to the processing of Sentinel-2 imagery and the generation of Satellite-derived Total Suspended Matter data for Lake Tagua Tagua. All authors contributed to the interpretation of results, critically revised the manuscript, and approved the final version.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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